

For these exercises, you should review the notes on [Neural Networks](#).

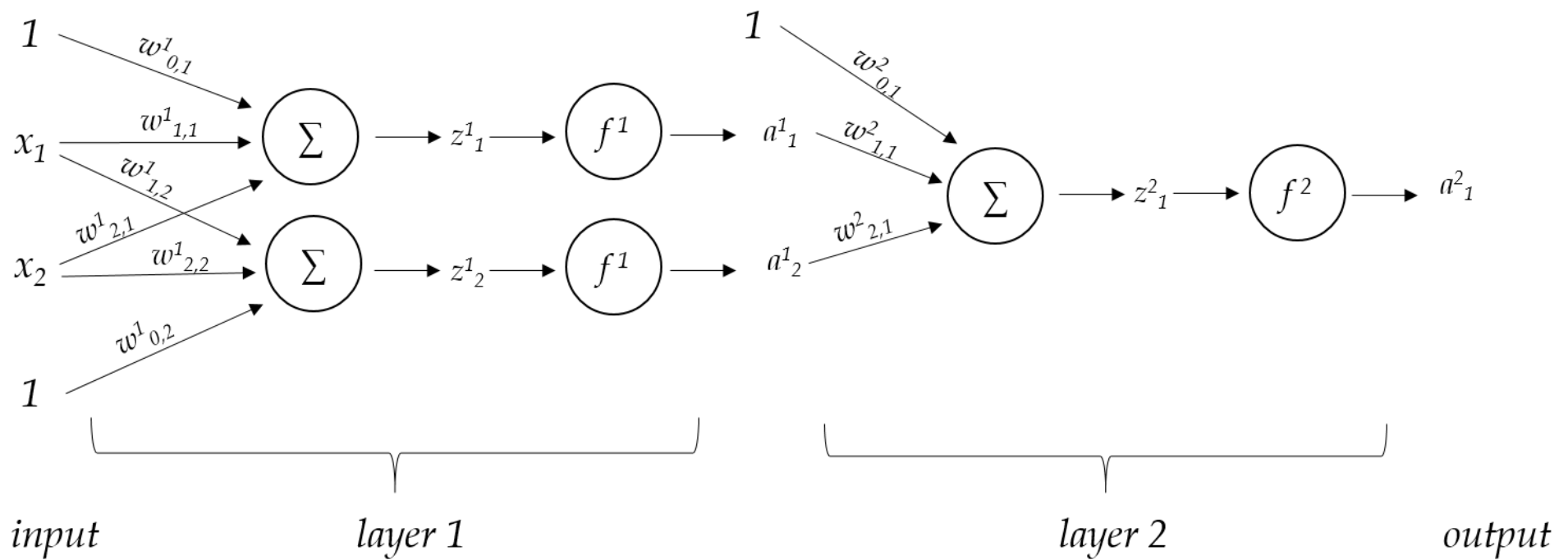
1) Prediction

Consider the following data set:

```
X = np.array([[0, 1, 2],  
              [0, 1, 2]])  
Y = np.array([[0, 1, 0]])
```

The columns of X and Y are the data points and corresponding labels.

We will be looking at the behavior of the following simple two-layer network:



Assume that within each layer, each unit has the step activation function $f(z)$ given by

$$f(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{otherwise} \end{cases}.$$

Let the weights in the first layer (layer 1) be:

- $w^1_{0,1} = -0.5$, $w^1_{1,1} = 1$, $w^1_{2,1} = 0$
- $w^1_{0,2} = 1.5$, $w^1_{1,2} = -1$, $w^1_{2,2} = 0$

1A) Enter a matrix Z where each column represents the outputs of the hidden units ($f^1(z_1^1)$ and $f^1(z_2^1)$) for each of the input vectors in X .

Enter a Python list of lists `[[a,b,c],[d,e,f]]`, each list is a row of the matrix.

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1B) Pick weights for the second layer $w_{0,1}^2, w_{1,1}^2, w_{2,1}^2$ so that the desired outputs are predicted correctly.

Enter a Python list of 3 numbers `[w0,12, w1,12, w2,12]`

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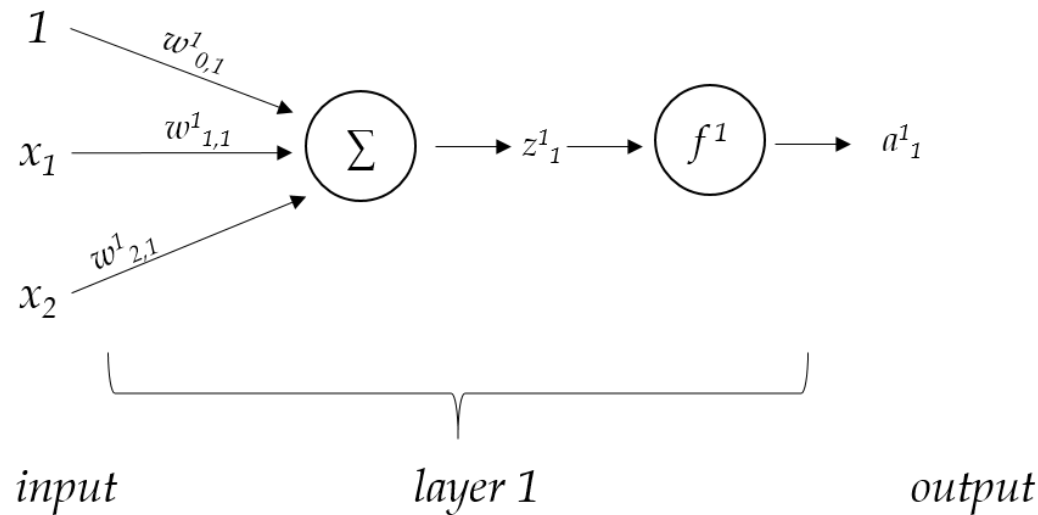
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2) Training

Now, we will consider the classification of a different set of X and Y data, with a different single layer network having the following structure and activation function:

$$z = w_{1,1}^1 x_1 + w_{2,1}^1 x_2 + w_{0,1}^1$$

In this network we have $f^1(z) = z$, so our output $a_1^1 = z_1^1$.



Assume the initial weights are $w_{0,1}^1 = 1$, $w_{1,1}^1 = 1$, $w_{2,1}^1 = 1$, and the step size is 0.5 (not usually a good idea, but okay for now).

The current training example is $x^{(i)} = [1, 2]^T$, $y^{(i)} = -1$.

2A) What is the output value a_1^1 , given current input $x^{(i)}$ and the current weights?

Enter a number

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2B) What will the values of weights $w_{0,1}^1, w_{1,1}^1, w_{2,1}^1$ be after one step of stochastic gradient descent at the given training example $x^{(i)} = [1, 2]^T, y^{(i)} = -1$ using our definition of [hinge loss](#) $L_h(v) = \max(0, 1 - v)$?

Enter a Python list of 3 numbers $[w_{0,1}^1, w_{1,1}^1, w_{2,1}^1]$ (to 3 decimal places)

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2C) What would the output value a_1^1 be, for this same input x , with these new weights?

Enter a number

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2D) What would happen to the v_i if we did another SGD update, for that same point, with step size 0.5, as before?

Enter a Python list of 3 numbers $[w_{0,1}^1, w_{1,1}^1, w_{2,1}^1]$

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2E) Now what would the output be?

Enter a number

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2F) What if we do one more update, for that same point?

Enter a Python list of 3 numbers $[w_{0,1}^1, w_{1,1}^1, w_{2,1}^1]$

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